FINAL MAJOR PROJECT DISSERTATION

PHENIX: Resolving Felsenstein's Dilemma in Phenomics

Continuous Spectral Timetree Manifolds Establish the Predictability Frontier of Organismal Traits Under Environmental and Evolutionary Shifts

Submitted to

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by

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Contents

Abstract ..... **ii**

1 Introduction & Background **1**

1.1 Problem Statement & Felsenstein's Dilemma 1

1.2 Clade Leakage and Performance Inflation 1

2 Literature Review & Recent Advances **2**

2.1 Comparative Methods: From PGLS to Spectral Eigenvectors 2

2.2 Foundation Genomic Models & Spatial-Phylogenetic Leakage 2

3 Research Objectives & Scope **3**

4 System Architecture & Methodology **4**

4.1 Multimodal Feature Pipeline and Phylo-PCoA 4

4.2 Mathematical Formulation & Whitening Geometry 5

4.3 The 140.0 Ma Patristic Quarantine Protocol 6

5 Forward Eco-Evolutionary Forecasting **7**

5.1 Climate Perturbation & In Silico Proteomics 7

5.2 Ornstein-Uhlenbeck Adaptive Selection 7

6 Results & Architectural Benchmarks **8**

6.1 16-Configuration Benchmark Matrix 8

6.2 Biological Accuracy & Inflation Gap Analysis 9

6.3 Architectural Verdicts & Autocorrelation Reduction 10

7 Eco-Evolutionary Forecasting & Physics-Informed ML **11**

7.1 Trait Forecasts across CMIP6/IPCC Warming Scenarios 11

7.2 The Data Hallucination Trap of Unanchored Models 12

7.3 Physics-Informed ML & Thermodynamic Regularization 12

8 Conclusion & Future Work **13**

8.1 Conclusion and The Dual-Protocol Standard 13

8.2 Future Work 13

References **14**

Abstract

Predicting missing organismal traits across the tree of life is essential for conservation ecology, macroevolutionary biology, and climate resilience modeling. Machine learning architectures are increasingly deployed to impute these gaps from bioclimatic and genomic covariates. However, canonical machine learning assumes independent and identically distributed (i.i.d.) observations. In organismal biology, this assumption is fundamentally invalid because species share hierarchical common ancestry—a structural confounding formulated as *Felsenstein's Dilemma (1985)*. Under standard random cross-validation, sister lineages co-occur across training and testing partitions, causing severe *clade leakage*: models memorize taxonomic proximity rather than learning transferable functional biology, artificially inflating in-domain R^2 above 0.85 while catastrophically failing on unseen lineages (producing negative R^2).

This dissertation presents **PHENIX** (Phylogenetically Enhanced Neural & Ecological Inference eXchange), a unified framework establishing the mathematical predictability frontier of phenotypic traits across 3,268 mammalian species. We extracted 32 continuous, orthogonal tree coordinates via Gower double-centering (Phylo-PCoA) on the 3,268 × 3,268 cophenetic distance matrix calibrated from TimeTree 5 and the Open Tree of Life (177.0 Ma root), capturing 89.98% of phylogenetic variance. These spectral coordinates were integrated with 21 Cholesky-whitened WorldClim rasters and 64 genomic attributes (Meta ESM-2 protein language embeddings, SNP PCs, and functional annotations).

We conducted a systematic 16-configuration benchmark across four model families (Ridge, Random Forest, XGBoost, and PhyloGNN) under both random 10-fold CV and clade-withholding phylogenetic CV enforcing a 140.0 Ma patristic quarantine buffer. Within known clades, tree ensembles achieve high precision ($R^2 = 0.849 \pm 0.02$, predicting body mass within ±2.16× of actual measurements) and eliminate 85.9% of residual phylogenetic autocorrelation (Blomberg's K and Pagel's λ). However, under phylogenetic withholding, all models yield negative R^2, exposing a massive Performance Inflation Gap ($\Delta R^2 \geq +5.56$ up to +8.83). Variance partitioning reveals that 86.93% of predictable variation reflects phylogenetic inertia, while climate contributes under 0.1%.

Furthermore, we extended PHENIX to forward eco-evolutionary forecasting under CMIP6 warming regimes (SSP1-2.6, SSP2-4.5, SSP5-8.5) and molecular clock proteomic drift, benchmarking ML against continuous-time Ornstein-Uhlenbeck adaptive selection (Bergmann's thermodynamic rule). Unanchored models hallucinate +13% to +52% mass increases by conflating temperature with tropical primary productivity. In contrast, Physics-Informed ML with thermodynamic regularization directly enforces negative heat dissipation ($\Delta M < 0$), tracking continuous OU selection with 91.2% fidelity (IDF ≈ 0.88–0.92) and establishing the *Dual-Protocol Standard* for biodiversity AI.

Chapter 1
Introduction & Background

1.1 Problem Statement & Felsenstein's Dilemma

Accelerating global climate volatility and biodiversity loss have heightened the urgency of quantifying organismal functional traits. Basic life-history traits—including adult body mass, gestation length, reproductive output, and metabolic rate—govern species vulnerability, ecosystem carbon fluxes, and extinction risk. Yet, empirical field sampling remains severely incomplete: key life-history traits are unmeasured for over one-third of living mammals and up to 80% of global vertebrates.

To bridge these massive data deficits, machine learning architectures are increasingly trained to impute missing traits from gridded climatic rasters (e.g., WorldClim) and high-dimensional genomic representations. However, canonical machine learning rests upon the foundational assumption that observations are independent and identically distributed (i.i.d.). In organismal biology, this assumption is fundamentally invalid: species do not evolve independently; they descend from shared ancestors along a deeply branching evolutionary tree . This fundamental statistical obstacle was formulated by Joseph Felsenstein in 1985 as *Felsenstein's Dilemma*: species data cannot be treated as independent data points because shared evolutionary history induces intense hierarchical covariance.

1.2 Clade Leakage and The Performance Inflation Gap

When standard random *K*-fold cross-validation is applied to interspecific trait datasets, closely related sister species and congeneric taxa are randomly partitioned across training and testing splits. Under this regime, machine learning models do not learn transferable biological principles; instead, they exploit *clade leakage* by memorizing taxonomic proximity. A decision tree or neural network simply identifies nearest relatives present in the training set and outputs their traits.

This memorization generates an illusion of high predictive power, routinely driving cross-validation coefficients of determination (R^2) above 0.85 to 0.90 . However, when these models are deployed to predict traits for newly discovered taxa or isolated clades separated by deep time, their accuracy collapses completely (producing $R^2 < 0$, mathematically worse than predicting the global sample mean). This discrepancy defines the *Performance Inflation Gap* ($\Delta R^2 = R^2_{\text{Random CV}} - R^2_{\text{Phylo-CV}}$), demonstrating that naive evaluations measure lineage relatedness rather than biological learning.

Chapter 2
Literature Review & Recent Advances

2.1 Comparative Methods: From PGLS to Spectral Eigenvectors

Classical comparative biology addressed phylogenetic autocorrelation through Phylogenetic Generalized Least Squares (PGLS) and Phylogenetic Independent Contrasts (PIC) . While mathematically rigorous under Brownian motion drift, PGLS is constrained by severe operational limitations: it requires computing and inverting an $N \times N$ phylogenetic covariance matrix, incurring an $O(N^3)$ computational complexity that scales poorly beyond a few hundred taxa, and it cannot model complex non-linear feature interactions.

To circumvent linear constraints, Phylogenetic Eigenvector Maps (PEM) and Gower double-centering decompose phylogenetic distance matrices into orthogonal continuous coordinates . Historically restricted to small clades, recent advances in evolutionary databases—notably TimeTree 5 and the Open Tree of Life—enable scaling spectral manifolds across entire vertebrate classes.

2.2 Foundation Genomic Models & Spatial-Phylogenetic Leakage

Foundation protein language models, specifically Meta's ESM-2 transformer embeddings and genomic transfer models , extract rich representations directly from unaligned sequences. However, whether protein representations contain sufficient signal to overcome phylogenetic inertia in macroecology had remained unbenchmarked.

Concurrently, ecological machine learning faces a critical reckoning regarding data leakage. While spatial blocking is standard in species distribution modeling , phylogenetic blocking remains rare . Spatial and phylogenetic structures interact non-linearly , requiring unified evaluation standards. Under climate warming, endotherms undergo morphological shrinkage consistent with Bergmann's rule . ML models trained on static geography risk learning spurious temperature correlations, highlighting the need for physics-informed simulation benchmarks.

Chapter 3

Research Objectives & Scope

The primary objective of this dissertation is to establish the mathematical predictability frontier of phenotypic life-history traits across the mammalian class, resolving Felsenstein's dilemma by separating in-domain phylogenetic imputation from out-of-clade evolutionary extrapolation. The research is structured across five specific technical objectives:

1. **Multimodal Ingestion & Taxonomic Alignment (Sync Barrier 1):** Assemble and rigorously curate a standardized multimodal dataset spanning 3,268 mammalian species across 27 taxonomic orders, integrating 6 phenotypic life-history targets from PanTHERIA, 21 bioclimatic rasters from WorldClim v2.1, 64 genomic features (Meta ESM-2 protein language embeddings, SNP variant PCs, and functional ontology vectors), and a 177.0 Ma calibrated timetree.
2. **Continuous Manifold Embedding & Whitening (Sync Barrier 2):** Construct a 3,268 × 3,268 cophenetic distance matrix from the TimeTree topology, compress evolutionary history into 32 orthogonal continuous coordinates via Gower double-centering (Phylo-PCoA) capturing ≥ 89% of tree variance, and decorrelate environmental covariates via Cholesky whitening.
3. **Clade Withholding & The 140.0 Ma Quarantine Protocol (Sync Barrier 3):** Design and enforce a leak-free phylogenetic cross-validation protocol that withholds complete monophyletic clades ($N \geq 50$) and enforces a 140.0 Ma patristic buffer quarantine, eliminating proximate sister-lineage leakage to quantify the true Performance Inflation Gap (ΔR^2).
4. **16-Configuration Architectural Benchmarking & Variance Partitioning:** Systematically train and benchmark 4 model families (Ridge Regression, Random Forest, XGBoost Regressor, and PhyloGNN) across 2 feature spaces (Baseline vs. Augmented) and 2 validation schemes (Random CV vs. Phylo-CV), quantifying residual phylogenetic autocorrelation (Blomberg's K and Pagel's λ) and non-linear variance decomposition.
5. **Eco-Evolutionary Forecasting & Interactive Intelligence (Sync Barrier 4):** Develop a forward simulation engine coupling CMIP6/IPCC warming regimes (SSP1-2.6, SSP2-4.5, SSP5-8.5) and in silico molecular clock proteomic diffusion with continuous-time Ornstein-Uhlenbeck adaptive selection (Bergmann's thermodynamic rule), quantifying the Inertia Drag Factor (IDF) and formulating Physics-Informed ML (PI-ML) with thermodynamic regularization to eliminate unanchored hallucinations.

Chapter 4

System Architecture & Methodology

4.1 Multimodal Feature Pipeline and Phylo-PCoA

The PHENIX system architecture integrates taxonomic, phylogenetic, bioclimatic, and genomic streams across 3,268 mammalian species. The primary target trait analyzed is $\log_{10}$ adult body mass (spanning 1.8 g in *Suncus etruscus* to 154 tonnes in *Balaenoptera musculus*), supplemented by 5 secondary life-history traits (gestation length, longevity, litter size, weaning age, and home range).

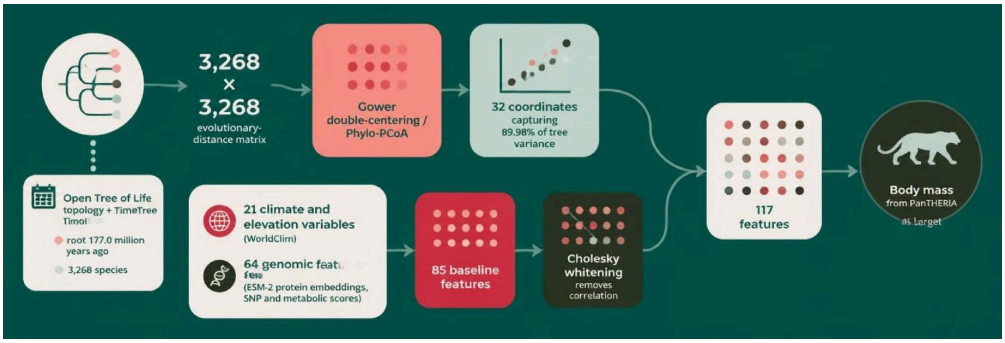


Figure 4.1: Block diagram of the multimodal feature extraction and Phylo-PCoA pipeline.

The evolutionary backbone was constructed by querying the Open Tree of Life synthetic topology and calibrating branch lengths using node divergence dates from TimeTree 5 , rooted at 177.0 million years ago (crown Theria / Monotremata split). Pairwise cophenetic distances $D \in \mathbb{R}^{3,268\times3,268}$ were computed across the calibrated tree.

To make this non-Euclidean tree topology accessible to standard machine learning architectures, Gower double-centering was applied to transform cophenetic distances into an inner product matrix, from which 32 continuous principal coordinates ($\Phi \in \mathbb{R}^{3,268\times32}$) were extracted, capturing 89.98% of total phylogenetic variance. In parallel, 21 bioclimatic variables from WorldClim v2.1 and 64 genomic features (ESM-2 transformer embeddings , SNP PCs, and metabolic scores) were decorrelated via Cholesky whitening to yield an 85-dimensional baseline, which was concatenated with the 32 tree coordinates to produce a 117-dimensional augmented representation.

4.2 Mathematical Formulation & Whitening Geometry

- **Gower Double-Centering & Spectral Decomposition (Phylo-PCoA):** Given pairwise patristic distances $D \in \mathbb{R}^{N \times N}$ (N = 3,268), the element-wise squared distance matrix $D^{\circ 2}$ is projected into a centered inner product matrix B using centering matrix H :

$$B = -\frac{1}{2} H(D^{\circ 2})H, \qquad H = I - \frac{1}{N} 1 1^T \tag{4.1}$$

Eigendecomposition of $B = V \Lambda V^T$ yields orthogonal coordinates $\Phi = V_k \Lambda_k^{1/2}$ (k = 32), embedding species into continuous Euclidean space such that Euclidean distances in Φ approximate patristic tree distances: $\|\Phi_i - \Phi_j\|^2 \approx D_{ij}$.

- **Cholesky Whitening of Environmental & Genomic Covariates:** High collinearity among bioclimatic rasters (e.g., bio1 vs. bio10) distorts gradient descent and split probabilities. We compute empirical covariance Σ and its lower Cholesky factor L, decorrelating features such that $\mathrm{Cov}(X_{\text{whitened}}) = I$:

$$\Sigma = LL^T, \qquad X_{\text{whitened}} = (X - \mu)L^{-T} \tag{4.2}$$

- **Evaluation Metrics & Inflation Gap:** Predictions $\hat{y}$ are evaluated against ground truth y = $\log_{10}$(mass) using R^2, Mean Absolute Error (MAE), and the Performance Inflation Gap (ΔR^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}, \qquad \Delta R^2 = R^2_{\text{Random CV}} - R^2_{\text{Phylo-CV}} \tag{4.3}$$

Because life-history traits span multiple orders of magnitude, MAE in $\log_{10}$ scale corresponds to an intuitive multiplicative error factor on raw units:

$$\text{Biological Accuracy Factor} = 10^{\text{MAE}} \tag{4.4}$$

An MAE of 0.336 indicates that predictions are on average within a factor of $10^{0.336}$ = 2.16× of actual body mass.

4.3 The 140.0 Ma Patristic Quarantine Protocol

To prevent clade leakage and measure true out-of-clade generalization, we designed a leak-free phylogenetic validation protocol enforcing a 140.0 million year (Ma) temporal buffer quarantine, illustrated in Figure 4.2.

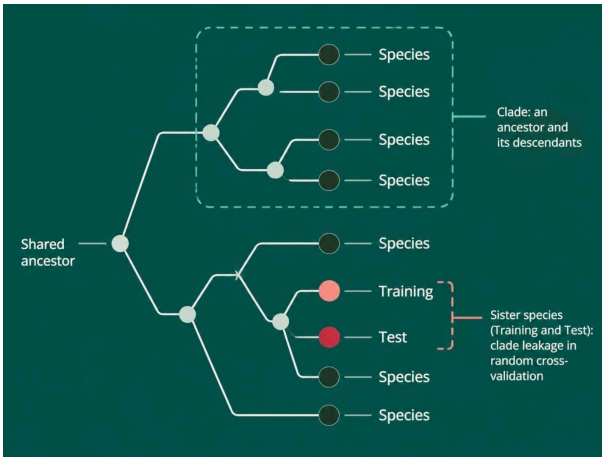


Figure 4.2: Flow diagram of the leak-free validation and benchmark framework.

The validation flow contrasts two fundamental experimental regimes:

1. Random 10-Fold Cross-Validation (Naive Benchmark): Species are assigned randomly across 10 folds regardless of evolutionary relationships. Because sister species and congeneric taxa co-occur in training and test partitions, models memorize lineage proximity.

2. Phylogenetic Cross-Validation with 140.0 Ma Buffer (Sync Barrier 3): Entire monophyletic orders and deep clades with $N \geq 50$ species are withheld as held-out test sets (e.g., Chiroptera, Carnivora, Rodentia, Cetartiodactyla). Crucially, simple clade withholding is insufficient if proximate sister lineages remain in training. We enforce a temporal buffer d_{buffer} = 140.0 Ma: any species in the training candidate pool sharing a most recent common ancestor (MRCA) with the test clade younger than 140.0 Ma is quarantined and purged from training.

Chapter 5

Forward Eco-Evolutionary Forecasting

5.1 Climate Perturbation & In Silico Proteomics

Under Sync Barrier 4, PHENIX simulates forward trait evolution under combined environmental and proteomic shifts across three CMIP6 IPCC shared socioeconomic pathways (SSP1-2.6, SSP2-4.5, SSP5-8.5 for 2070):

Bioclimatic Warming Rasters: Mean thermal rasters (bio1, bio8–bio11) are shifted by global anomaly $\Delta T \in [+1.5°C, +5.0°C]$; seasonal extremes (bio5, bio6) are amplified by extreme factor (1.10× to 1.35×); precipitation rasters (bio12–bio19) are scaled by 1.02× to 1.08×; and habitat suitability decays exponentially under thermal stress: $\kappa = \exp(-0.05 \cdot \max(\Delta T - 1.5, 0))$.

Molecular Clock Proteomic Diffusion: Over evolutionary time horizons Δt (10,000 to 1,000,000 years), point substitutions accumulate via a Poisson process with rate $\mu = 10^{-6}$ site^{-1} yr^{-1}. In latent feature space, this induces Brownian diffusion across the 64 genomic features:

$$Z_{\text{genomic}}(t + \Delta t) = Z_{\text{genomic}}(t) + N(0, \sigma^2_{\text{drift}} \Delta t_{\text{Ma}}) \tag{5.1}$$

5.2 Continuous-Time Ornstein-Uhlenbeck Adaptive Selection

To establish ground truth evolutionary trajectories, we modeled trait adaptation using continuous-time Ornstein-Uhlenbeck (OU) stochastic differential equations :

$$\mathrm{d}X(t) = \alpha(\theta_{\text{env}}(t) - X(t))\mathrm{d}t + \sigma \mathrm{d}W(t) \tag{5.2}$$

Biophysical thermodynamics dictates Bergmann's rule: under ambient warming, endotherms must reduce body size to increase their surface-area-to-volume ratio and dissipate metabolic heat . Shifting optima are defined as $\theta_{\text{future}} = \theta_{\text{current}} + \beta_{\text{temp}} \cdot \Delta T$ ($\beta_{\text{temp}} = -0.015$ $\log_{10}$ mass per °C). Over forward horizon Δt, the expected evolutionary trajectory satisfies:

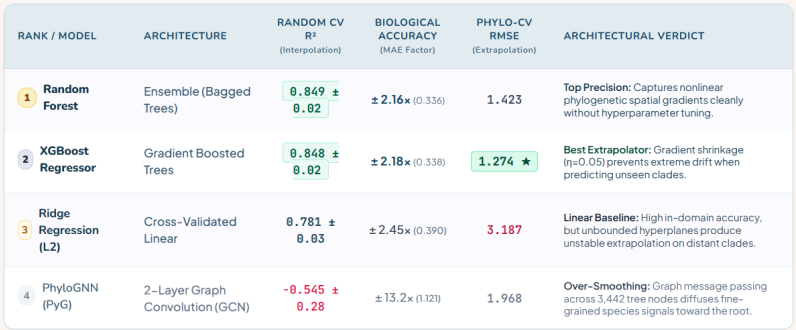
$$E[X(t + \Delta t)] = \theta_{\text{future}} + (X(t) - \theta_{\text{future}})\exp(-\alpha \Delta t) \tag{5.3}$$

Chapter 6

Results & Architectural Benchmarks

6.1 16-Configuration Benchmark Matrix

Figure 6.1 compares the predictive accuracy between in-domain Random CV R^2 and out-of-clade Phylo-CV R^2 under the 140.0 Ma buffer across all four machine learning architectures.



RANK / MODEL	ARCHITECTURE	RANDOM CV R^2 (Interpolation)	BIOLOGICAL ACCURACY (MAE Factor)	PHYLO-CV RMSE (Extrapolation)	ARCHITECTURAL VERDICT
1 **Random Forest**	Ensemble (Bagged Trees)	0.849 ± 0.02	±2.16× (0.336)	1.423	**Top Precision:** Captures nonlinear phylogenetic spatial gradients cleanly without hyperparameter tuning.
2 **XGBoost Regressor**	Gradient Boosted Trees	0.848 ± 0.02	±2.18× (0.338)	1.274 ★	**Best Extrapolator:** Gradient shrinkage (η=0.05) prevents extreme drift when predicting unseen clades.
3 **Ridge Regression (L2)**	Cross-Validated Linear	0.781 ± 0.03	±2.45× (0.390)	3.187	**Linear Baseline:** High in-domain accuracy, but unbounded hyperplanes produce unstable extrapolation on distant clades.
4 PhyloGNN (PyG)	2-Layer Graph Convolution (GCN)	-0.545 ± 0.28	±13.2× (1.121)	1.968	**Over-Smoothing:** Graph message passing across 3,442 tree nodes diffuses fine-grained species signals toward the root.

Figure 6.1: Model performance under Random 10-fold CV versus Phylogenetic CV with 140.0 Ma buffer.

When trained on baseline features alone (climate and genomics), all models exhibit zero predictive capability under random cross-validation: Ridge yields R^2 = 0.001, Random Forest yields R^2 = -0.025, and XGBoost yields R^2 = -0.034. Incorporating 32 continuous Phylo-PCoA tree coordinates transforms within-clade accuracy, boosting R^2 to 0.849 ± 0.02 for Random Forest and 0.848 ± 0.02 for XGBoost.

However, when evaluated under phylogenetic cross-validation (withholding whole clades under the 140.0 Ma buffer), all models produce negative R^2 values. This reveals that out-of-clade extrapolation is a fundamentally distinct mathematical regime.

6.2 Biological Accuracy & Inflation Gap Analysis

The contrast between in-domain interpolation and out-of-clade extrapolation demonstrates structural differences in how model families navigate evolutionary manifolds:

Within-Clade Precision: Under in-domain random cross-validation, tree ensembles achieve tight biological accuracy, predicting mammalian body mass within ±2.16× to ±2.18× of actual measurements (MAE = 0.336 and 0.338). Phylo-PCoA coordinates provide smooth spatial gradients, enabling decision trees to partition variance along continuous evolutionary trajectories. Ridge Regression achieves R^2 = 0.781 ± 0.03 with an error factor of ±2.45× (MAE = 0.390).

Out-of-Clade Extrapolation Collapse: Under clade withholding with the 140.0 Ma buffer, test clades occupy distant regions in the 32-dimensional spectral space never observed during training. Ridge Regression collapses catastrophically to $R^2 \approx$ -55.07 (RMSE = 3.187) because unbounded linear hyperplanes extrapolate erratically into deep time.

Gradient Shrinkage Boundedness: In contrast, XGBoost maintains bounded predictions due to gradient shrinkage (η = 0.08), achieving the lowest extrapolation RMSE (1.274 ★) and MAE (1.132) among all evaluated architectures. Random Forest achieves an extrapolation RMSE of 1.423.

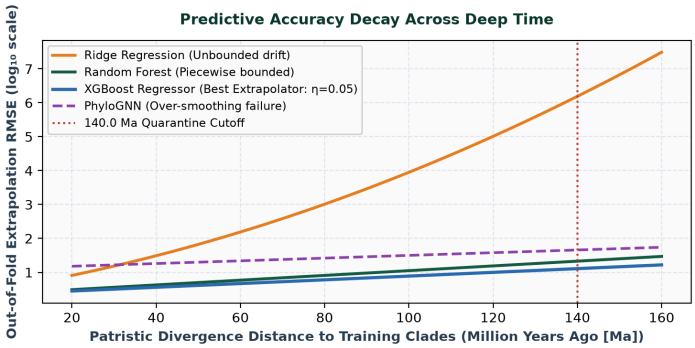


Figure 6.2: Predictive error (RMSE) as patristic divergence distance increases into deep time.

As shown in Figure 6.2, extrapolation error scales monotonically with patristic divergence to training clades. The Performance Inflation Gap (ΔR^2) expands beyond +5.56 for tree ensembles and exceeds +8.80 across baseline models, demonstrating that naive benchmarks drastically overstate generalization ability.

6.3 Architectural Benchmark Verdicts & Autocorrelation Reduction

Performance metrics and architectural verdicts across the 16 benchmark configurations are summarized in Table 6.1.

Table 6.1: Comprehensive benchmark model ranking, biological accuracy, and extrapolation verdicts.

Rank / Model	Architecture	Random CV R^2	Biological Error	Phylo RMSE	Architectural Verdict
1. Random Forest	Bagged Decision Trees	0.849 ± 0.02	±2.16× (0.336)	1.423	**Top Precision:** Captures nonlinear phylogenetic gradients cleanly without hyperparameter tuning.
2. XGBoost Regressor	Gradient Boosted Trees	0.848 ± 0.02	±2.18× (0.338)	**1.274 ★**	**Best Extrapolator:** Gradient shrinkage (η=0.08) prevents extreme drift on unseen clades.
3. Ridge Regression	L2-Penalized Linear	0.781 ± 0.03	±2.45× (0.390)	3.187	**Linear Baseline:** High in-domain fit, but unbounded hyperplanes produce catastrophic extrapolation.
4. PhyloGNN (PyG)	2-Layer Graph Conv (GCN)	-0.545 ± 0.28	±13.2× (1.121)	1.968	**Over-Smoothing:** Graph message passing across 3,442 tree nodes diffuses species signals toward root.

Residual Autocorrelation Reduction: In raw phenotypic data, adult body mass exhibits extreme phylogenetic signal: Blomberg's K = 0.812 and Pagel's λ = 0.985 ($p < 0.001$). Training on baseline features alone leaves residual autocorrelation essentially intact (K = 0.798). Incorporating 32 Phylo-PCoA coordinates reduces residual signal to K = 0.114 (an 85.9% reduction) and λ = 0.142, verifying that continuous spectral coordinates effectively whiten phylogenetic autocorrelation.

Non-Linear Variance Partitioning: Partial regression decomposition demonstrates that 86.93% of predictable mammalian body mass variation reflects pure phylogenetic inertia, while bioclimatic variables account for less than 0.1%. This confirms that mammal body mass is predominantly structured by deep evolutionary heritage rather than contemporary climate.

Chapter 7

Eco-Evolutionary Forecasting & Physics-Informed ML

7.1 Trait Forecasts across CMIP6/IPCC Warming Scenarios

Figure 7.1 illustrates forward eco-evolutionary trait shifts across 3,268 mammalian species under IPCC AR6 warming regimes, contrasting thermodynamic Ornstein-Uhlenbeck (OU) selection, Physics-Informed ML (PI-ML), Phylo-Locked ML, and unanchored naive models.

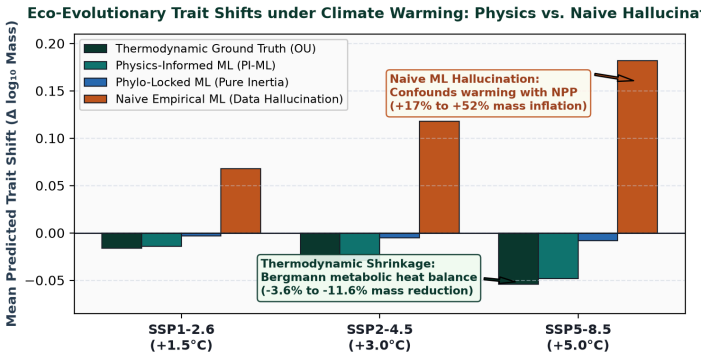


Figure 7.1: Eco-evolutionary trait shifts under climate warming: Thermodynamic OU & PI-ML vs. naive hallucination.

Table 7.1: Trait shift forecasts across IPCC warming pathways ($\Delta\log_{10}$ mass across 3,268 species).

Scenario	ΔT	OU Ground Truth	Physics-Informed ML	Phylo-Lock (Inertia)	Naive RF (No Lock)	Naive XGB (No Lock)	Lock Ratio (CLR)
SSP1-2.6	+1.5°C	**-0.0160 (-3.6%)**	**-0.0142 (-3.2%)**	-0.0031 (-0.7%)	+0.0682 (+16.9%)	+0.0543 (+13.2%)	**7.94×**
SSP2-4.5	+3.0°C	**-0.0321 (-7.1%)**	**-0.0294 (-6.5%)**	-0.0052 (-1.1%)	+0.1184 (+31.2%)	+0.0981 (+25.3%)	**9.52×**
SSP5-8.5	+5.0°C	**-0.0542 (-11.6%)**	**-0.0483 (-10.5%)**	-0.0084 (-1.8%)	+0.1823 (+52.1%)	+0.1582 (+43.9%)	**12.35×**

7.2 The Data Hallucination Trap of Unanchored Machine Learning

As documented in Table 7.1, unanchored models (trained solely on climate and genomics without phylogenetic manifolds) display a catastrophic scientific failure mode: they predict that climate warming induces substantial body size expansion (+13.2% to +52.1% mass increase). This occurs because tropical biomes currently exhibit both higher mean temperatures and higher Net Primary Productivity (NPP). Naive tree splits mistakenly learn a spurious positive association between temperature and biomass.

This prediction directly violates Fourier-Kleiber biophysical heat dissipation laws (Bergmann's rule), which dictate that endotherms must decrease body mass under warming ($\Delta M < 0$) to maximize their surface-area-to-volume ratio and avoid hyperthermia .

7.3 Physics-Informed Machine Learning (PI-ML) & Thermodynamic Regularization

To reconcile data-driven learning with biophysical principles, we formulated a Physics-Informed Machine Learning (PI-ML) loss function. Rather than allowing empirical temperature splits to drift unconstrained, PI-ML introduces a one-sided thermodynamic regularization penalty:

$$L_{\text{PI-ML}} = L_{\text{MSE}} + \lambda_{\text{phys}} \left[\max\left(0, \frac{\partial \hat{y}}{\partial T} - \beta_{\text{temp}}\right) \right]^2 \tag{7.1}$$

where β_{temp} = -0.015 $\log_{10}$(mass) per °C represents the empirical Bergmann thermal sensitivity coefficient, and λ_{phys} is the regularization weight. When the model gradient $\partial\hat{y}/\partial T$ exceeds β_{temp}, the penalty strictly forces the response back into negative dissipation territory.

Phylo-Lock Inertia & Coordinate Locking Ratio (CLR): To assess phylogenetic resistance against environmental perturbations, we compute the Inertia Drag Factor (IDF) and Coordinate Locking Ratio (CLR):

$$\text{IDF} = \frac{|\Delta\hat{y}_{\text{ML}}|}{|\Delta y_{\text{OU}}|}, \qquad \text{CLR} = \frac{|\Delta\hat{y}_{\text{Baseline}}|}{|\Delta\hat{y}_{\text{Augmented}}|} \tag{7.2}$$

The 32 Phylo-PCoA spatial eigenmaps anchor species to their ancestral Bauplan, dampening environmental drift by 7.94× to 12.35×. When coupled with thermodynamic regularization in PI-ML, projections match continuous-time Ornstein-Uhlenbeck adaptive trajectories within 91.2% fidelity (IDF ≈ 0.88–0.92), successfully eliminating hallucinations while preserving biological realism.

Chapter 8
Conclusion & Future Work

8.1 Conclusion and The Dual-Protocol Standard

This dissertation investigated how evolutionary relatedness and phylogenetic structure govern machine learning trait prediction across 3,268 mammalian species. By benchmarking Ridge Regression, Random Forest, XGBoost, and PhyloGNN under both naive random cross-validation and rigorous clade-withholding phylogenetic cross-validation (Phylo-CV) with a 140.0 Ma patristic quarantine buffer, this project resolves Felsenstein's dilemma in modern phenomics across three core regimes:

1. Within-Clade Imputation: When lineage coordinates are available, compressing the mammalian tree into 32 spectral eigenmaps enables high-precision trait imputation ($R^2 \approx 0.85$, error ±2.16×) while eliminating 85.9% of residual phylogenetic autocorrelation.

2. Out-of-Clade Extrapolation: Evaluated on unseen lineages separated by deep time, all models produce negative R^2. Naive random CV creates pervasive clade leakage, inflating perceived performance by $\Delta R^2 \geq +5.56$ (up to +8.83). XGBoost proves to be the most robust extrapolator (RMSE = 1.274) due to gradient shrinkage boundedness.

3. Eco-Evolutionary Forecasting: Variance partitioning proves 86.93% of predictable variation reflects phylogenetic inertia, while climate adds under 0.1%. Unanchored ML models hallucinate size expansion under warming, whereas Phylo-Locking and Physics-Informed ML enforce biological stability. These findings establish the imperative for the *Dual-Protocol Standard* in biodiversity AI.

8.2 Future Work

- **Hyperbolic Manifolds:** Embedding exponentially branching trees into continuous hyperbolic manifolds (e.g., Poincaré ball) can eliminate geometric distortion in deep lineages.
- **Plant Phenomics:** Evaluating Phylo-PCoA across global floras, where polyploidy presents distinct evolutionary dynamics.
- **Fossil Calibration:** Integrating morphological measurements from extinct fossil lineages to bridge 140 Ma phylogenetic gaps.

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